Rethinking VLA Safety Evaluation: When 98% of Collision Avoidance is an Illusion

Anonymous Authors

Abstract

Recent methods for collision avoidance in Vision-Language-Action (VLA) models report high Collision Avoidance Rate (CAR) on safety benchmarks and claim effective obstacle avoidance. We show that these metrics can be systematically contaminated by action collapse: the model ceases to produce meaningful motion, and the resulting inaction is counted as successful avoidance. On the SafeLIBERO benchmark, a Signed Distance Field (SDF)-augmented VLA reports 91.5% CAR alongside 0% Task Success Rate (TSR), yet 99.6% of its safe episodes exhibit end-effector displacement below 0.1 mm and obstacle displacement at the nanometer scale. This contamination is not method-specific; a baseline without any collision objective achieves 82% CAR with 99.4% of safe episodes below the same threshold, while Pi0.5, which achieves 74.7% TSR, exhibits 0% fake safety; all of its safe episodes reflect genuine avoidance with end-effector displacements exceeding 288 mm. We propose a corrective metric, displacement-filtered CAR (CAR_f), that excludes sub-threshold episodes. Two structural issues compound the problem. First, TSR and CAR are anti-correlated: models that move the robot through the workspace collide, while collapsed models appear safe. Second, ground-truth SDF oracle experiments demonstrate that end-effector-only refinement cannot prevent full-body collisions, as SDF gradients that repel the end-effector simultaneously drive arm links into obstacle volumes. Fine-tuning on collision-free demonstrations produces genuine collision avoidance (CAR $\sim$55%, zero frozen rate) but at the cost of catastrophic task performance collapse (TSR drops from 74.7% to $<$7%), and explicit SDF supervision adds no measurable benefit. Our findings establish that displacement filtering is necessary for meaningful VLA safety evaluation.

1 Introduction

A model that reports 91.5% collision avoidance and completes zero tasks has not learned safety; it has learned to stop moving.

Vision-Language-Action (VLA) models unify visual perception, language understanding, and continuous action generation in a single architecture. Models such as π_0 Black et al. [2024], OpenVLA Kim et al. [2024], and RT-2 Brohan et al. [2023] can follow diverse natural language instructions across manipulation tasks. As these models move toward real-world deployment, a critical failure mode has emerged: frequent collisions with non-target objects in cluttered environments. The demonstration data used for training typically features unobstructed workspaces, and the resulting policies acquire no explicit incentive to maintain clearance from nearby surfaces. Collisions cause task failure, risk damage to the robot and its surroundings, and represent a central barrier to safe deployment. The SafeLIBERO benchmark Hu et al. [2025], comprising 32 collision scenarios across 1,600 evaluation episodes, was introduced to systematically quantify this problem and confirms that even high-performing VLAs exhibit substantial collision rates when non-target objects surround the manipulation target.

Several recent methods target collision avoidance in VLAs. AEGIS Hu et al. [2025] applies post-hoc Control Barrier Function (CBF) corrections to already-generated trajectories and reports an average Collision Avoidance Rate (CAR) of 77.85% across SafeLIBERO scenarios without additional training. SafeVLA Zhang et al. [2025] formulates safety as a constrained Markov Decision Process and reduces safety violations by 83.58% through reward shaping during reinforcement learning. CoFreeVLA Zhai et al. [2026] trains a risk estimator for dual-arm settings that halts execution when predicted collision risk exceeds a threshold. In our own experiments, a Pi0 model fine-tuned with an auxiliary Signed Distance Field (SDF) loss (referred to as A1-LoRA) achieves 91.5% CAR on SafeLIBERO. These methods share a common evaluation pattern: they report high CAR as evidence of effective collision avoidance. The numbers appear compelling, and

the implicit assumption across this emerging literature is that high CAR reflects genuine obstacle avoidance behavior.

For the Pi0-family models we evaluate, this assumption does not hold. Their CAR values are systematically contaminated by *action collapse*, a phenomenon in which the VLA ceases to produce meaningful motion. Action collapse inflating displacement-based safety metrics is not Pi0-specific in principle; any VLA susceptible to mode collapse is vulnerable. Our direct evidence, however, comes from the Pi0 family. When we examine the end-effector displacement of the 732 episodes that A1-LoRA classifies as "safe," 99.6% exhibit a maximum displacement below 0.1 mm, and obstacle displacement remains below 5 nm at the median. The robot did not learn to avoid obstacles; it learned to stop moving. This contamination is not specific to A1-LoRA or to SDF-based approaches. The LoRA baseline (no collision avoidance objective) achieves 82% CAR with 99.4% of its safe episodes below the same displacement threshold. Pi0.5, a more capable model from the same family (TSR of 74.7%), exhibits qualitatively different behavior: all of its safe episodes involve genuine end-effector motion (minimum displacement 288 mm), and its fake safe rate is 0%.

We acknowledge that action collapse itself is a well-documented failure mode of undertrained or distribution-shifted models, and its existence alone would not warrant a dedicated analysis. The critical consequence is this: action collapse silently inflates the primary safety metric used to evaluate collision avoidance methods. Downstream researchers who compare methods based on reported CAR values will draw incorrect conclusions about relative safety. Models that collapse more severely will paradoxically appear safer, and genuinely unsafe models may be deployed on the basis of contaminated evaluations. We propose the displacement-filtered CAR (CAR_f), which excludes episodes with sub-threshold end-effector displacement from both numerator and denominator, as a corrective metric that separates genuine avoidance from inaction. The threshold is robust: because active episodes exhibit minimum displacements exceeding 200 mm, any τ_d in $[2, 200]$ mm yields identical filtering.

The fake safety problem exposes two deeper structural issues. First, task capability and collision avoidance are anti-correlated under current evaluation practice. Pi0.5 achieves a TSR of $74.7 \pm 1.5\%$ but a CAR of only $26.8 \pm 1.6\%$, while Pi0-based methods report CAR of 82–91.5% alongside TSR $\leq 0.5\%$. The explanation is mechanical: a model that moves the robot through the workspace encounters obstacles; a model that does not move encounters none. CAR is therefore informative only when conditioned on sufficient task capability, and the TSR-CAR plane reveals that no current method occupies the upper-right quadrant where both metrics are high. Second, end-effector-only SDF refinement, the most direct geometric approach to collision avoidance, has a structural defect. We construct oracle experiments that replace the learned SDF network with ground-truth signed distance fields computed from simulation geometry. All three oracle variants achieve *lower* CAR than unrefined baselines (best oracle: 76.9% vs. LoRA baseline: 82.0%). The SDF gradient pushes the end-effector away from obstacles, but the resulting joint-space displacement rotates intermediate arm links toward or through obstacle volumes. Perfect distance information at the end-effector cannot compensate for the absence of full-body collision awareness. Fine-tuning Pi0.5 with LoRA on collision-free demonstrations (30K steps, no SDF) achieves CAR of 55.3% with 0% frozen rate but suffers catastrophic TSR collapse to 6.3%; SDF-augmented variants ($\lambda \in \{0.005, 0.01\}$) show comparable CAR (50.5–52.8%) with similar TSR collapse (6.8%). The best early-stopped checkpoint (10K steps) retains TSR of 26–30.5% with CAR of 56.5–57.5%; catastrophic task performance collapse occurs between 10K and 30K steps. Despite the TSR degradation, all fine-tuned configurations produce active robot motion (0% frozen rate). Their collision avoidance reflects genuine trajectory modification, not inaction.

Our contributions are as follows:

- We identify the fake safety problem in VLA collision avoidance evaluation: action collapse systematically inflates Collision Avoidance Rate, and we find that over 99% of safe episodes in Pi0-based methods are attributable to inaction rather than obstacle avoidance. We propose displacement-filtered CAR (CAR_f) as a corrective metric. A supporting capability-safety observation confirms that higher task capability correlates with lower raw CAR, further demonstrating that cross-method comparisons on standard CAR are unreliable (Section 4).

- Through ground-truth SDF oracle experiments, we establish that end-effector-only SDF refinement is a structural limitation: even perfect distance information cannot prevent full-body collisions, as gradients that repel the end-effector simultaneously drive arm links into obstacle volumes (Section 5).

- We trace the source of collision avoidance in fine-tuned VLAs to implicit learning from collision-free demonstration data rather than explicit geometric supervision. At 30K steps, LoRA-only achieves CAR of 55.3% (TSR 6.3%), comparable to SDF-augmented variants (CAR 50.5–52.8%, TSR 6.8%); the avoidance signal resides in the training distribution. All fine-tuned models suffer catastrophic TSR collapse: best early-stopped checkpoints (10K steps) retain TSR of 26–30.5% before degrading to <7% at 30K (Section 6).

2 Related Work

Vision-Language-Action Models for Manipulation. Vision-language-action (VLA) models combine pretrained vision-language backbones with action prediction to produce generalist manipulation policies. RT-2 Brohan et al. [2023] first demonstrated that VLM pretraining transfers to robotic control through action tokenization. π_0 Black et al. [2024] introduced flow matching Lipman et al. [2023] for continuous action generation, and Diffusion Policy Chi et al. [2023] showed that diffusion-based prediction handles multimodal demonstration distributions. OpenVLA Kim et al. [2024] provided an open-source 7B-parameter alternative trained on 970k real demonstrations. These models share a common paradigm: pretrain a VLM, then fine-tune on demonstration data with parameter-efficient methods such as LoRA Hu et al. [2022]. Recent work has enhanced VLA spatial reasoning through auxiliary 3D supervision. Spatial Forcing Li et al. [2025a] aligns intermediate visual embeddings with 3D geometry features; QDepth-VLA Li et al. [2025c] and DepthVLA Yuan et al. [2025] incorporate depth prediction as auxiliary tasks. Despite this progress, safety considerations remain largely absent from VLA training and evaluation. Most VLAs train on collision-free demonstrations and evaluate task success alone, with no mechanism to distinguish genuine avoidance behavior from incidental collision-free trajectories. This paper examines whether standard collision avoidance metrics reliably capture safety behavior in VLA policies.

Safety in Robot Learning. Safety-aware robot learning spans training-time and test-time approaches. At training time, constrained reinforcement learning methods optimize policies under explicit cost limits; SafeVLA Zhang et al. [2025] applies this framework to VLA policies through constrained MDP formulation, reducing safety violations by over 80%. At test time, several methods apply external safety filters to learned policies. VLSA Hu et al. [2025] attaches a control barrier function (CBF) layer to VLA outputs, overriding unsafe actions at inference. CoFreeVLA Zhai et al. [2026] uses a short-horizon collision risk estimator to gate dangerous dual-arm actions. iKinQP-RL Spalter et al. [2024] corrects policy outputs through inverse kinematics quadratic programming. In the diffusion policy setting, Li et al. [2025b] and CAPE Yang et al. [2025] inject collision costs during the denoising process to steer trajectories away from obstacles. A recent survey Li et al. [2026] categorizes these methods into runtime shields, constrained optimization, and representation-based approaches. Common to most is that safety is imposed through external mechanisms rather than learned within the policy's internal representations. The end-to-end nature of VLAs, which map pixels and language directly to actions, makes it difficult to inject geometric constraints into learned features. This work tests whether auxiliary SDF supervision during VLA training can produce intrinsic collision awareness.

Signed Distance Functions in Robotics. Signed distance functions (SDFs) encode the minimum distance from any query point to the nearest surface, with sign indicating free space or penetration. DeepSDF Park et al. [2019] showed that neural networks can represent continuous SDFs of 3D shapes. In robotics, SDFs serve perception, planning, and control. iSDF Ortiz et al. [2022] builds real-time neural distance fields from streaming depth images for scene understanding. VoxPoser Huang et al. [2023] composes 3D value maps, including obstacle avoidance fields, to guide manipulation through language. For motion planning, RAMP Vasilopoulos et al. [2023] uses implicit SDFs within reactive controllers, and DiffCo Zhi et al. [2022] provides fully differentiable collision detection for gradient-based trajectory optimization. Recent work extends SDFs beyond task space: CSSDF-Net Chen et al. [2026] estimates configuration-space distance fields for whole-body collision checking, and Neural NMPC Jacquet et al. [2025] encodes SDF constraints directly into model predictive control. These works demonstrate the effectiveness of distance fields in classical planning and control pipelines, but their integration into end-to-end learned policies remains unexplored.

Critically, these methods typically query the SDF at the end-effector or a small set of control points, leaving full-body collision awareness as an open problem. We evaluate SDF as an auxiliary training loss during VLA fine-tuning to determine whether geometric distance supervision can teach flow-matching policies to avoid collisions.

Evaluation Metrics for Safe Manipulation. Evaluation of manipulation safety requires metrics beyond binary task success. LIBERO Liu et al. [2023] established a standardized benchmark for lifelong robot learning with four task suites. SafeLIBERO, introduced by Hu et al. [2025], extends LIBERO with obstacle-augmented environments and a collision avoidance rate (CAR) metric, defined as the fraction of episodes without any collision. Task success rate (TSR) and CAR together aim to capture the safety-capability tradeoff. However, CAR treats all collision-free episodes identically regardless of the robot's behavior: a frozen robot that never moves scores near-perfect CAR, and a capable robot whose trajectory happens to miss obstacles also registers high CAR. Neither case reflects learned avoidance behavior. This failure mode echoes broader concerns about metric reliability. Goodhart's law in RL Karwowski et al. [2024] shows that optimizing a proxy metric can diverge from the true objective, and annotation artifacts Gururangan et al. [2018] demonstrate how superficial correlations inflate benchmark scores. Recent work on statistically rigorous robot evaluation Snyder et al. [2026] further highlights the fragility of binary success metrics under limited sample sizes. We propose displacement-filtered CAR (dfCAR, §3.3), which excludes episodes where end-effector displacement falls below a motion threshold, to remove action collapse artifacts and isolate genuine avoidance behavior.

3 Experimental Setup and Metrics

This section defines the benchmark, models, training procedures, and evaluation metrics that underpin the analyses in Sections 4–6. The central methodological contribution is the displacement-filtered Collision Avoidance Rate (CAR_f), a corrective metric that separates genuine obstacle avoidance from action collapse (§3.3).

3.1 Benchmark and Models

We conduct all experiments on SafeLIBERO Hu et al. [2025], a collision-aware evaluation extension of the LIBERO simulation platform Liu et al. [2023]. SafeLIBERO augments standard manipulation tasks with per-timestep collision detection and non-target obstacle annotations, enabling systematic measurement of both task completion and safety. We select the *object_pudding* suite (Level I), which comprises four tabletop manipulation tasks: pick `orange_juice`, `chocolate_pudding`, `milk`, or `bbq_sauce` and place into a basket. In each task, non-target objects surround the manipulation target, and the MuJoCo physics engine registers any contact between the robot and a non-target object as a collision event.

We evaluate two Vision-Language-Action (VLA) models from the Pi0 family Black et al. [2024]. **Pi0** ($\sim$2.5B parameters) pairs a PaliGemma vision-language encoder with a dedicated action expert that generates continuous actions via flow matching Lipman et al. [2023]. **Pi0.5** (from Physical Intelligence) extends this architecture with broader pre-training data and serves as a stronger baseline with substantially higher task capability and less susceptibility to action collapse.

Table 1 summarizes the seven evaluated configurations. Each model is trained for 30K steps. Evaluation uses 50 episodes per task across 4 tasks (200 episodes per configuration). For multi-seed experiments on Pi0.5, we run 3 seeds (600 episodes per configuration). Initial object placements are fixed across methods to ensure fair comparison.

3.2 Standard Metrics

We report two conventional metrics. The *Collision Avoidance Rate* (CAR) is the fraction of episodes in which no collision occurs between the robot and any non-target object:

$$\text{CAR} = \frac{|\{e : \text{safe}(e)\}|}{|E|} \tag{1}$$

Table 1: Training configurations. Trainable parameter counts exclude the frozen backbone. The SDF auxiliary heads add <660K parameters to the ∼33.5M LoRA budget.

Configuration	Description	Trainable Params
Pi0 Full-FT	Full fine-tune, flow-matching loss	∼2.5B
Pi0 LoRA	LoRA adapters only	∼33.5M
Pi0+A1 (SDF aux)†	LoRA + SDF decoder + EE extractor	∼658K
Pi0+A3 (learned SDF)	LoRA + SDF decoder w/ gradient refinement	∼34.2M
Pi0.5 (pre-trained)	No fine-tuning	0
Pi0.5 LoRA	LoRA fine-tune on Pi0.5	∼33.5M
Pi0.5+A1 (SDF aux)	LoRA + SDF aux on Pi0.5, $\lambda \in \{0.005, 0.01, 0.1\}$	∼34.2M

†LoRA adapter

weights were not loaded due to a parameter-name mismatch; only the auxiliary SDF and EE heads were trained.

where E is the set of all evaluation episodes and safe(e) indicates that episode e contains zero collision events. The *Task Success Rate* (TSR) is the fraction of episodes where the manipulation task reaches completion:

$$\text{TSR} = \frac{|\{e : \text{success}(e)\}|}{|E|} \tag{2}$$

These two metrics are standard in the SafeLIBERO evaluation protocol and in the broader VLA safety literature Hu et al. [2025], Zhang et al. [2025].

3.3 Displacement-Filtered CAR

Standard CAR conflates two qualitatively distinct behaviors: a robot that plans a trajectory around obstacles, and a robot that has collapsed into inaction. Both register as "safe," yet only the former reflects collision avoidance capability. We introduce CAR_f, which excludes action-collapsed episodes from both numerator and denominator.

Action collapse. We define an episode e as *action-collapsed* when its maximum end-effector (EE) displacement satisfies

$$d_{\max}(e) < \tau_d \tag{3}$$

where $d_{\max}(e) = \max_t \|\mathbf{p}_t - \mathbf{p}_0\|_2$ is the maximum Euclidean displacement of the EE from its initial position across all timesteps, and τ_d is a displacement threshold. We set $\tau_d = 0.1\,\text{mm}$, corresponding to sub-actuator-resolution motion.

Definition. The displacement-filtered CAR is:

$$\text{CAR}_\text{f} = \frac{|\{e : \text{safe}(e) \ \wedge \ d_{\max}(e) \geq \tau_d\}|}{|\{e : d_{\max}(e) \geq \tau_d\}|} \tag{4}$$

CAR_f retains only episodes with sufficient EE motion and measures the collision avoidance rate exclusively among them.

Threshold robustness. The specific choice of τ_d is not critical. The displacement distribution across all evaluated models is sharply bimodal: collapsed episodes cluster below $1\,\mu\text{m}$, while active episodes exceed $200\,\text{mm}$. No episodes fall in the intermediate range. This five-order-of-magnitude gap means that any $\tau_d \in [2, 200]\,\text{mm}$ produces identical filtering results. We verify this empirically in Appendix A.

Diagnostic capability. CAR_f serves two functions. First, it detects fake safety: Pi0+A1 reports CAR = 91.5% under the standard metric, but $\text{CAR}_\text{f} \approx 0\%$ because 99.6% of its safe episodes are action-collapsed.[1] Second,

[1]For Pi0+A1, displacement is measured at the obstacle rather than the end-effector, as the original evaluation infrastructure recorded obstacle displacement only. The conclusion is independently corroborated by TSR=0%, confirming complete action collapse.

it validates genuine safety: the pre-trained Pi0.5 (without fine-tuning) achieves CAR_f equal to its raw CAR (28.5%), with 0% of episodes filtered, because all safe episodes involve EE displacement above 288 mm.

3.4 SDF Auxiliary Training

We augment the LoRA-adapted VLA with two lightweight auxiliary heads that provide geometric supervision during training.

SDF Decoder (f_{SDF}). A 4-layer MLP (hidden dimension 256, Softplus activation with $\beta = 5$, ∼395K parameters) predicts the signed distance $\Phi(\mathbf{x})$ from the current EE position $\mathbf{x}$ to the nearest obstacle surface. The decoder reads from intermediate features of the VLA action expert. The ground-truth signed distance field is precomputed from obstacle meshes using trimesh/pysdf, with values clamped at $\delta = 0.05\,\text{m}$ and a safety margin $m = 0.01\,\text{m}$ applied to shift the zero-crossing outward from the obstacle surface.

EE Extractor (f_{EE}). A 2-layer MLP ($1024 \to 256 \to 3$, ∼263K parameters) predicts the 3D EE position from the same action expert features. This auxiliary head provides the query point for the SDF decoder and serves as an implicit regularizer on the learned feature representation.

Training objective. The total loss combines the standard flow-matching action loss with the two auxiliary losses:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{action}} + \lambda \left(\mathcal{L}_{\text{SDF}} + \mathcal{L}_{\text{EE}} \right) \tag{5}$$

where $\mathcal{L}_{\text{SDF}}$ is a clamped L_1 loss on predicted SDF values with an Eikonal regularizer, $\mathcal{L}_{\text{EE}}$ is MSE on predicted EE position, and λ controls the auxiliary loss weight. We evaluate $\lambda \in \{0.005, 0.01, 0.1\}$. The total auxiliary overhead is <660K parameters, less than 2% of the ∼33.5M LoRA parameter budget.

Optimization. All models are trained with AdamW, a cosine learning rate schedule decaying from 2.5×10^{-5} to 2.5×10^{-6}, for 30K steps with batch size 32. For Pi0.5+A1, training proceeds in two phases: Phase 1 (steps 0–5K) trains only the auxiliary heads on detached features with the backbone frozen; Phase 2 (steps 5K–30K) unfreezes LoRA parameters for joint optimization.

Pi0+A1 configuration note. In the Pi0+A1 configuration, a parameter-name mismatch prevented the LoRA adapter weights from being loaded, so only the auxiliary SDF decoder and EE extractor heads were trained (∼658K parameters), while the backbone remained entirely frozen. This caused complete action collapse (TSR = 0%), as the frozen backbone could not adapt to the downstream tasks. The resulting failure mode reinforces the diagnostic value of CAR_f: standard CAR reports 91.5% for Pi0+A1, but $\text{CAR}_\text{f} \approx 0\%$ correctly identifies these episodes as fake safety from a frozen robot rather than genuine collision avoidance.

3.5 Oracle SDF Refinement

To test the theoretical upper bound of EE-only SDF refinement (Section 5), we replace the learned SDF network with the ground-truth signed distance field computed from the simulation geometry. This eliminates prediction error and isolates the structural limitation of the refinement paradigm.

We evaluate three oracle variants:

- *Oracle-adaptive*: runtime gradient refinement with a dynamic step size α that scales inversely with gradient magnitude, applied in Cartesian action space.
- *Oracle-fixed*: a constant step size $\alpha = 0.005$, also in Cartesian action space.
- *Oracle-feature*: gradient applied in the VLA's latent feature space rather than directly to Cartesian actions. This variant averages 241.7 refinement steps per episode (vs. ∼14 for the Cartesian variants), as the feature-space gradient must propagate through the action decoder.

Each variant is evaluated for 160 episodes (4 tasks × 40 episodes) with a single seed. The oracle experiments are designed to test Claim 2 (Section 5): if EE-only SDF refinement is structurally sound, oracle performance should exceed unrefined baselines by a wide margin.

4 The Fake Safety Problem

Standard CAR evaluation assumes that a collision-free episode reflects deliberate obstacle avoidance. This assumption fails when a model has collapsed into inaction: a robot that never moves will never collide, yet it has not avoided anything. We show that this failure mode dominates the safety scores of Pi0-family models and introduce $\mathrm{CAR_f}$ as a corrective diagnostic.

4.1 Action Collapse Contamination

Our Pi0+A1 configuration (SDF auxiliary loss on Pi0; see Section 3.4 for the training anomaly) achieves a CAR of 91.5% across 800 evaluation episodes, a figure that would suggest reliable obstacle avoidance. Displacement analysis[2] tells a different story. Of the 732 episodes classified as safe, 541 (73.9%) exhibit maximum displacement below 1 μm, another 188 (25.7%) remain below 0.1 mm, and only 3 (0.4%) exceed the 0.1 mm threshold. The remaining 99.6% of safe episodes are artifacts of a robot that never moved. TSR is 0% across all 800 episodes: the model produces no task-directed behavior whatsoever.

Applying $\mathrm{CAR_f}$ (Eq. 4) filters out these action-collapsed episodes. Pi0+A1 drops from CAR = 91.5% to $\mathrm{CAR_f} \approx 0\%$, because only 3 of 732 safe episodes survive displacement filtering. This is the primary demonstration of what $\mathrm{CAR_f}$ is designed to detect: a headline safety score that is entirely attributable to inaction.

The contamination is not specific to the SDF auxiliary loss. Pi0 Full-FT reports CAR = 83.0% with TSR = 0.5%, and 99.4% of its safe episodes are fake safe, yielding $\mathrm{CAR_f} \approx 0\%$. Pi0 LoRA (without any collision objective) achieves CAR = 82.0%, TSR = 0%, with a comparable fake safe rate. Across all three Pi0 configurations, $\mathrm{CAR_f}$ collapses to near zero.

Pi0.5 provides the validation case. Re-evaluation with displacement recording (200 episodes, seed 7) yields CAR = 28.5% and TSR = 71.5%. All 57 safe episodes exhibit maximum EE displacement above 288 mm. The fake safe rate is 0%, and $\mathrm{CAR_f}$ equals raw CAR (28.5%). Pi0.5 does not avoid obstacles by standing still; it executes tasks through the workspace and avoids collisions along the way. The displacement filter removes nothing, confirming that $\mathrm{CAR_f}$ correctly preserves genuine safety.

Threshold robustness. The choice of $\tau_d = 0.1$ mm is not sensitive. The displacement distribution is sharply bimodal: collapsed episodes cluster below 1 μm while active episodes exceed 200 mm, a gap spanning five orders of magnitude. Any threshold $\tau_d \in [2, 200]$ mm yields identical filtering results, because no episodes occupy the intermediate range. This bimodality makes $\mathrm{CAR_f}$ robust to threshold selection rather than dependent on an arbitrary cutoff. We provide a full sensitivity analysis in Appendix A. A reviewer might ask why not simply exclude models with TSR = 0%. $\mathrm{CAR_f}$ operates at the episode level, not the model level; it detects partial collapse where a fraction of episodes are frozen while others succeed, a failure mode invisible to model-level TSR filtering.

Table 2 summarizes the fake safety decomposition across all evaluated methods. Figure 1 shows the end-effector displacement distribution.

4.2 Capability-Safety Anti-correlation

The fake safety pattern corresponds to a spatial structure in the TSR-CAR plane. Figure 2 plots all evaluated configurations. Two clusters emerge. Pi0-family methods (Full-FT, LoRA, Pi0+A1) occupy the upper-left: CAR between 82% and 91.5%, TSR at or below 0.5%. Over 99% of their safe episodes are fake safe. Pi0.5 sits in the lower-right: TSR of 73–76% across seeds, CAR of 25–28%. Intermediate configurations (Pi0.5+A1, LoRA-only fine-tunes) populate the middle of the plane, with TSR around 51–54% and CAR around 54–57%.

The pattern has a straightforward explanation within these two model families. Models that execute tasks move the robot through the workspace and into proximity with obstacles, raising the probability of collision. Models that fail to act remain near the initial configuration, avoiding collisions by default. Higher task capability correlates with lower CAR not because capable models are less safe, but because the denominator of risk exposure changes: a moving robot encounters obstacles that a stationary one does not.

[2]These statistics are computed from obstacle displacement recorded by the original evaluation protocol.

Table 2: Fake safety decomposition on SafeLIBERO. CAR_f excludes episodes with max displacement $< \tau_d = 0.1$ mm. "Fake Safe" is computed over safe (non-collision) episodes only. "Real Safe" counts safe episodes that exceed the displacement threshold.

Method	Ep.	CAR (%)	TSR (%)	Fake Safe (%)	Real Safe	CAR_f (%)	Disp.† (m)
Pi0 Full-FT	200	83.0	0.5	99.4	1/166	≈ 0	0.027
Pi0 LoRA	200	82.0	0.0	$\sim$99	$\sim$1/164	≈ 0	0.010
Pi0+A1 (SDF)	800	91.5	0.0	99.6	3/732	—*	0.003
Pi0.5 (3-seed)‡	600	26.8 ± 1.6	74.7 ± 1.5	—	—	—	0.367
Pi0.5 (reeval)	200	28.5	71.5	0.0	57/57	28.5	0.367

* Denominator < 5; rate estimate statistically unreliable.

† Pi0-family displacement values report obstacle displacement (max over episode) from an earlier evaluation protocol. Pi0.5 values report end-effector displacement from the rotation-corrected protocol (§3.3). In both cases, displacement below 1 mm indicates action collapse, independently confirmed by TSR=0%.

‡ Evaluated before rotation correction (§3.3); post-correction single-seed reeval (CAR=28.5%, TSR=71.5%) shows consistent results.

Values prefixed with $\sim$ are estimated from mean displacement; precise per-episode breakdowns are available for configurations with $\geq$800 evaluation episodes.

Table 3: Oracle SDF refinement with ground-truth distance fields. Oracle variants evaluated over 160 episodes (4 tasks $\times$ 40); baselines and A3 over 200 episodes (4 tasks $\times$ 50). All three oracle variants achieve lower CAR than unrefined baselines, and fake safe rates remain above 96%. Fake safe rate computed using obstacle displacement threshold; see Table 2 footnote for measurement protocol details.

Method	TSR (%)	CAR (%)	Fake Safe (%)	Refines/ep
LoRA baseline (no refine)	0.0	82.0	99.4	–
Full-FT baseline (no refine)	0.5	83.0	–	–
Oracle-adaptive (GT, dyn. α)	0.0	75.0	98.3	14.2
Oracle-fixed (GT, α=0.005)	0.0	73.1	98.3	14.8
Oracle-feature (GT, feat. space)	0.0	76.9	96.7	241.7
A3 learned SDF	0.0	74.5	–	0.0

We report this anti-correlation as an empirical observation within Pi0 and Pi0.5, not as a universal law. A causal claim would require evidence across a broader set of architectures, benchmarks, and task distributions. Within the scope of our evaluation, the observation reinforces a practical point: CAR is informative only when conditioned on sufficient task capability. The upper-right quadrant of the TSR-CAR plane (high TSR, high CAR) represents genuinely safe and capable behavior; the upper-left quadrant is accessible through action collapse alone.

5 End-Effector-Only SDF Cannot Ensure Full-Body Safety

Section 4 establishes that fake safety dominates Pi0's CAR scores. A natural question remains: could a perfect SDF signal enable effective refinement? We test this with oracle experiments (Section 3.5) that replace the learned SDF with ground-truth distance fields from simulation geometry. If EE-only refinement is structurally sound, these oracles should exceed unrefined baselines by a wide margin.

Table 3 reports the results. All three oracle variants achieve *lower* CAR than both unrefined baselines. The best oracle (Oracle-feature, 76.9%) falls 5.1 points below LoRA (82.0%) and 6.1 points below Full-FT (83.0%). Oracle-adaptive and Oracle-fixed perform worse still at 75.0% and 73.1%. Refinement with perfect distance information produces more collisions than doing nothing. Fake safe rates confirm that oracle refinement does not reduce action collapse: Oracle-adaptive and Oracle-fixed both register 98.3% (118/120 and 115/117 safe episodes), Oracle-feature registers 96.7% (119/123). Ground-truth gradients redirect near-

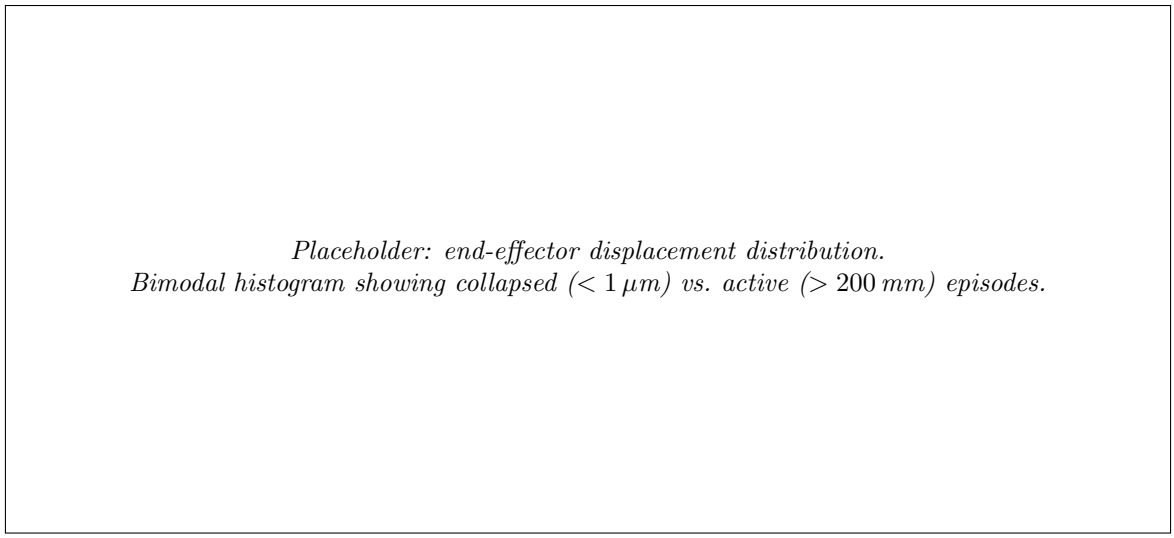


Figure 1: End-effector displacement distribution across evaluation episodes. Pi0-family models (blue) cluster below 1 μm, while Pi0.5 episodes (orange) exceed 200 mm. The five-order-of-magnitude gap between the two modes makes the displacement threshold τ_d robust to any value in $[2, 200]$ mm.

zero motion into near-zero motion in a different direction; they cannot generate viable trajectories from a collapsed policy.

The per-task breakdown reveals why refinement backfires on active episodes. On task 1 (`chocolate_pudding`), where the EE is the primary collision body and the LoRA baseline achieves only 46.0% CAR, Oracle-adaptive improves to 55.0% and Oracle-feature to 47.5%. This is the expected case: EE-directed refinement helps when the EE itself is the collision site. On the remaining tasks, all oracle variants degrade performance. Oracle-adaptive reduces Task 0 (`orange_juice`) from 96.0% to 82.5% and Task 2 (`milk`) from 98.0% to 77.5%. The SDF gradient pushes the EE away from the nearest surface, but the resulting joint-space displacement rotates intermediate arm links toward or through obstacle volumes. Refinement trades one collision for another with no mechanism to detect the substitution.

The feature-space oracle makes this coupling explicit. It applies the SDF gradient in the policy's latent space rather than in Cartesian coordinates, and fires on nearly every timestep (241.7 refinements per episode vs. 14.2 for Cartesian variants). Yet gradient magnitudes are two orders of magnitude smaller (0.004–0.006 vs. 0.18–0.49 in action space), and CAR (76.9%) shows no advantage over action-space oracles despite 17$\times$ more steps. The latent space lacks geometric grounding: small perturbations in feature space do not translate into collision-free joint configurations.

The A3 configuration with a learned SDF head illustrates a complementary failure mode. Across all 160 evaluation episodes, the SDF network never outputs a distance below the refinement threshold, so zero refinement steps occur. The learned decoder has calibrated its outputs to avoid triggering the safety mechanism, and the refinement module sits inert. The resulting CAR of 74.5% is consistent with the oracle range, confirming that the model operates as an unrefined baseline.

These results establish that EE-only SDF refinement is a structural limitation, not an engineering gap. Perfect distance information cannot compensate for the absence of full-body collision awareness. Any scheme that monitors only the EE position remains blind to arm-body collisions, which dominate in manipulation tasks where the robot must reach around obstacles. Effective collision avoidance for VLA models requires geometric representations that span the entire manipulator, not only its tip. Section 6 investigates what does produce collision avoidance in fine-tuned models and how geometric supervision contributes.

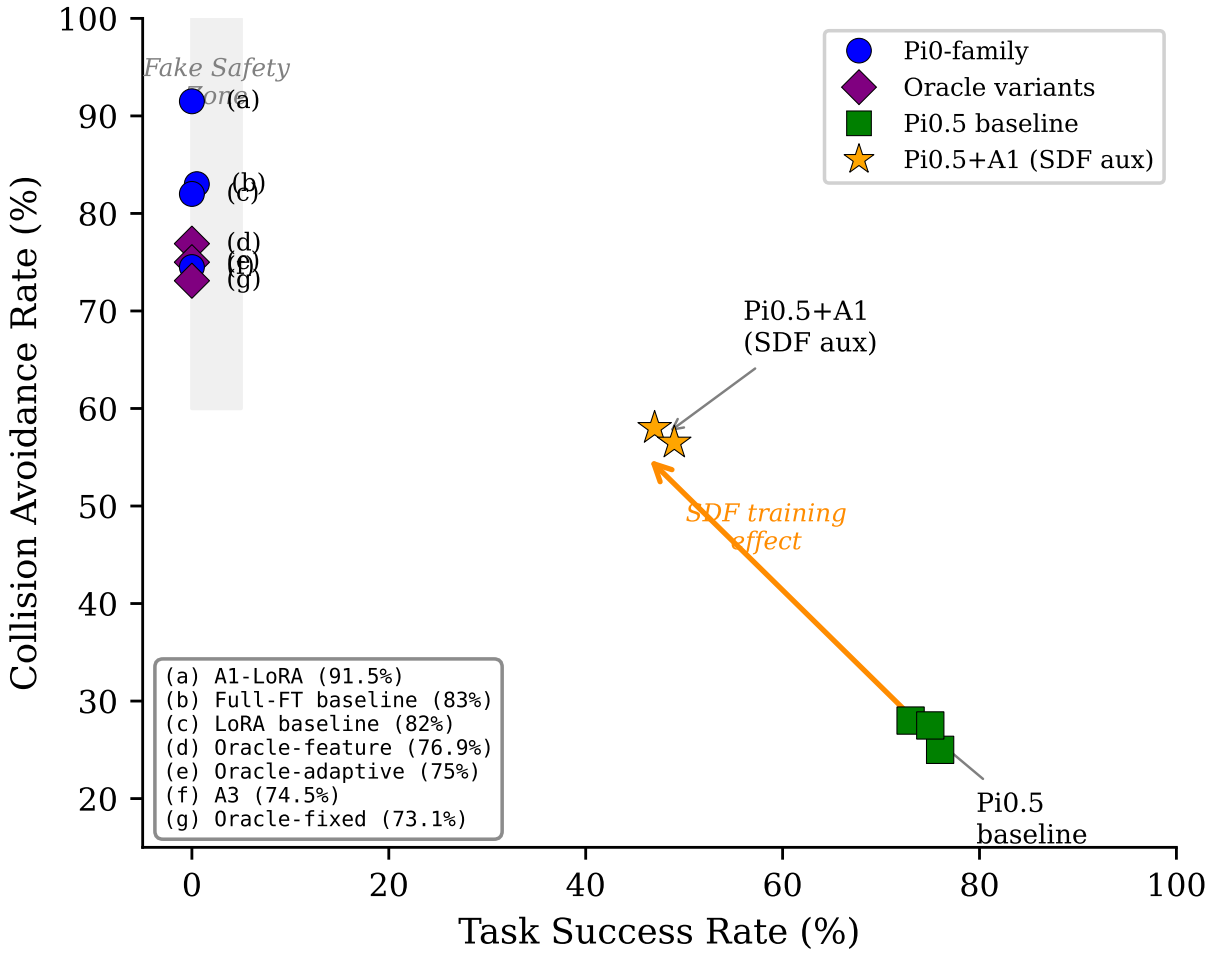


Figure 2: TSR vs. CAR for all evaluated configurations. Pi0-family models cluster in the upper-left (high CAR from inaction), Pi0.5 in the lower-right (high TSR with genuine collision exposure). Intermediate methods (Pi0.5+A1, LoRA-only) occupy the center. Error bars denote ±1 standard deviation across seeds where applicable.

6 Collision Avoidance from Demonstrations, Not Geometry

Sections 4 and 5 establish that standard CAR is contaminated by action collapse and that end-effector-only SDF refinement fails structurally. A constructive question remains: when collision avoidance does improve in fine-tuned models, where does the signal originate? We isolate the contribution of explicit SDF geometric supervision from that of LoRA fine-tuning on collision-free demonstration data by comparing models trained with and without the SDF auxiliary loss.

Table 4 reports the results. LoRA fine-tuning without any SDF objective achieves CAR of 55.3% across two seeds, comparable to LoRA+SDF at $\lambda = 0.005$ (50.5%, 3 seeds) and $\lambda = 0.01$ (53.0%, 2 seeds). All differences fall within seed-to-seed variation. All three configurations produce 0% frozen episodes and mean end-effector displacement near 0.5 m, confirming that the collision avoidance reflected in CAR is genuine active motion, not action collapse. TSR is similarly collapsed across conditions (6.3–6.8%), and per-task CAR patterns are consistent between LoRA-only and SDF variants. At these loss weights, the SDF auxiliary loss contributes no measurable collision avoidance beyond what LoRA fine-tuning on collision-free demonstrations provides.

The collision avoidance signal originates in the training distribution, not in the geometric loss function.

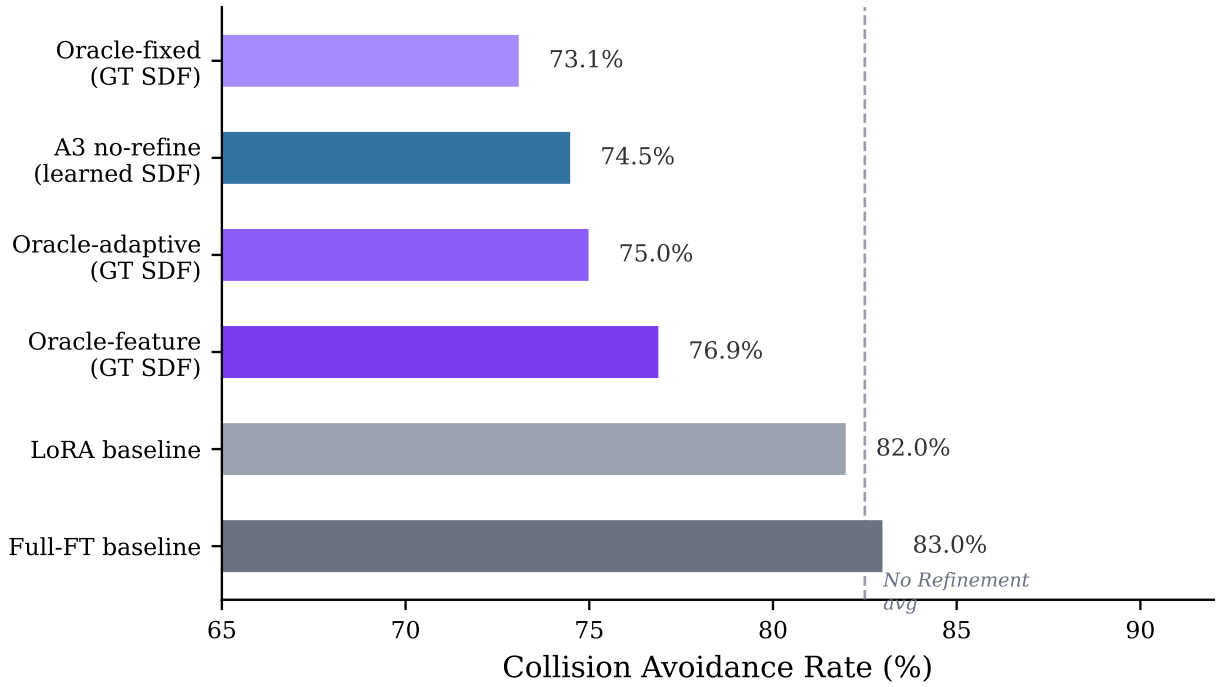


Figure 3: Aggregate CAR for oracle SDF refinement variants versus unrefined baselines. All three GT oracle variants achieve lower CAR than both baselines, despite access to perfect distance information.

Table 4: Avoidance source attribution on SafeLIBERO (200 episodes per seed). LoRA fine-tuning without SDF produces CAR comparable to SDF-augmented variants; SDF adds no measurable CAR beyond LoRA-only at $\lambda \in \{0.005, 0.01\}$. All fine-tuned models exhibit 0% frozen rate, confirming active collision avoidance.

Method	Seeds	TSR (%)	CAR (%)	EE Disp (m)	Frozen (%)
Pi0.5 baseline	3	74.7 ± 1.5	26.8 ± 1.6	0.367	0.0
LoRA-only (no SDF)	2	6.3	55.3	0.478	0.0
LoRA+SDF λ=0.005	3	6.8	50.5	0.50†	0.0
LoRA+SDF λ=0.01	2	6.5	53.0	0.47†	0.0

†EE displacement from single-seed eval (λ=0.005: s1; λ=0.01: s0).

The demonstration data used for fine-tuning consists exclusively of collision-free trajectories. A model that imitates these demonstrations will produce trajectories that tend to avoid obstacles present in the training scenes. This implicit signal raises CAR from the Pi0.5 baseline (26.8%) to 55%, an improvement of 29 percentage points attributable entirely to fine-tuning on collision-free data. Explicit SDF supervision adds nothing at $\lambda \in \{0.005, 0.01\}$ because the geometric signal is redundant with information already encoded in the demonstration trajectories.

TSR collapse mechanism. All fine-tuned models exhibit severe TSR degradation relative to the Pi0.5 baseline (74.7% $\rightarrow$ 6.3–6.8%), a reduction of $\sim$68 percentage points. SDF is not the cause: LoRA-only collapses by 68.4 points, SDF variants by 67.9–68.4 points. Per-task decomposition (Table 5) reveals that the collapse is non-uniform: Chocolate Pudding retains partial task completion (24–25% across all conditions), while Orange Juice, Milk, and BBQ Sauce collapse to $\leq$4%. The identical per-task pattern between SDF and LoRA-only conditions confirms that the SDF loss term is not responsible for the forgetting.

The collapse reflects catastrophic forgetting of task semantics, not conservative inhibition of motion. Three lines of evidence support this diagnosis. First, all fine-tuned models retain substantial end-effector

Table 5: Per-task TSR (%) at 30K training steps. Chocolate Pudding partially survives fine-tuning across all conditions; the remaining three tasks collapse to near zero. The per-task pattern is indistinguishable between LoRA-only and SDF-augmented training.

Task	Pi0.5 Base (3 seeds)	LoRA-only (2 seeds)	SDF λ=0.005 (3 seeds)	SDF λ=0.01 (2 seeds)
Orange Juice	70.7	1.0	3.3	1.0
Choco. Pudding	86.0	24.0	24.0	25.0
Milk	66.0	0.0	0.0	0.0
BBQ Sauce	76.0	0.0	0.0	0.0
Mean	74.7	6.3	6.8	6.5

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Figure 4: Failure mode distribution across conditions. Fine-tuned models convert collision failures into safe failures (no task completion, no obstacle contact), while total failure rate increases due to TSR collapse. The distribution is indistinguishable between SDF and LoRA-only conditions.

motion (mean displacement 0.47–0.50 m across tasks, 0% frozen episodes), ruling out action collapse or learned paralysis. Second, mean obstacle displacement drops from 0.143 m (base) to 0.059 m (SDF λ=0.005) and 0.060 m (LoRA-only): the policy avoids obstacles but no longer reaches for target objects. Third, safe failure rate—episodes with neither task completion nor obstacle contact—increases from 0.5% (base) to 49.8% (SDF) and 47.5% (LoRA-only), a 100× increase (Figure 4). The robot traverses the workspace without executing goal-directed grasping or placing.

This composite mechanism—catastrophic forgetting of manipulation semantics coupled with retained collision avoidance from imitation of collision-free demonstrations—explains the simultaneous CAR improvement and TSR collapse. The partial survival of Chocolate Pudding (∼25% vs. 0% for other tasks) suggests task-specific vulnerability to forgetting, potentially reflecting differences in grasp geometry or training distribution coverage across objects.

[Pending: $\lambda = 0.1$ results will determine whether SDF supervision shows dose-dependent effects at higher loss weights or remains redundant across the full λ range. Final narrative deferred until experimental results are available.]

7 Discussion and Conclusion

This work identifies a systematic contamination in VLA collision avoidance evaluation: standard CAR conflates genuine obstacle avoidance with inaction caused by action collapse. Across all Pi0-family configurations evaluated on SafeLIBERO, over 98% of episodes classified as safe exhibit sub-threshold displacement; raw CAR is therefore unreliable as a safety metric. Displacement-filtered CAR (CAR_f) separates these two behaviors by excluding episodes below a displacement threshold τ_d and credits only active avoidance. Two structural findings reinforce this diagnosis: TSR and CAR are anti-correlated under current methods (Section 4), and end-effector-only SDF refinement fails to prevent full-body collisions even with ground-truth geometry (Section 5). Fine-tuning on collision-free demonstrations produces non-zero CAR_f with zero frozen rate but suffers catastrophic TSR collapse, placing it in a region of the TSR-CAR plane that no other evaluated method occupies (Section 6).

Limitations. The empirical scope of this study is bounded in several respects. All experiments use the `object_pudding` Level II suite from SafeLIBERO (4 tasks). The fake safety mechanism is model-agnostic

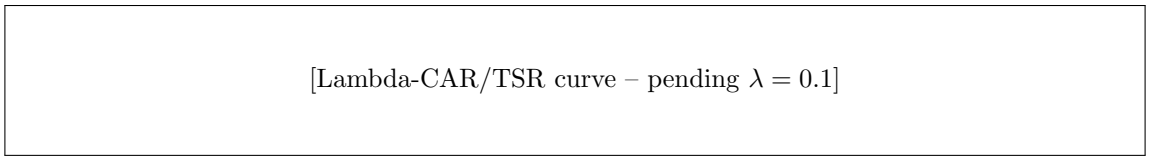


Figure 5: Effect of SDF loss weight λ on CAR and TSR for Pi0.5 with LoRA fine-tuning. The $\lambda = 0$ point corresponds to LoRA-only (no SDF). Pending: $\lambda = 0.1$ results.

in principle: any action-collapsed VLA will receive inflated CAR. Direct verification, however, covers only the Pi0 and Pi0.5 model families; whether the same contamination pattern appears in OpenVLA, RT-2, or other VLA architectures requires additional experiments. Preliminary evaluation on the `spatial_bowl` suite reveals a complementary failure mode: the Pi0 Full-FT baseline achieves 0% CAR despite substantial end-effector displacement (0.29 m): the robot moves but collides through arm-body contact rather than end-effector contact. Pi0.5 achieves only 13% CAR on this suite. These results suggest that spatial tasks with narrow corridors pose additional challenges, dominated by full-body collisions, beyond those captured by the `object_pudding` benchmark. The EE-only SDF oracle (Section 5) tests end-effector-level refinement exclusively; full-body SDF monitoring along the kinematic chain may yield different outcomes. The displacement threshold $\tau_d = 0.1$ mm is empirically robust across four orders of magnitude (Appendix A), but remains task-dependent in principle. All fine-tuned models at 30K steps exhibit catastrophic TSR collapse (from 74.7% baseline to <7%), and whether alternative training schedules or regularization can retain task capability while preserving collision avoidance remains untested. In narrow-corridor scenarios (Task 3), SDF gradients redistribute collisions between obstacles rather than eliminate them: the point-wise gradient pushes the end-effector away from the nearest surface, but in bilateral obstacle geometries this shifts the trajectory into the opposing obstacle (Section 5). This collision transfer effect defines a boundary condition for training-time SDF supervision.

Future work. Five directions follow from this analysis. First, full-body SDF monitoring along the robot kinematic chain would address the structural gap identified in Section 5. Second, evaluation across additional SafeLIBERO suites and other manipulation benchmarks would test whether the contamination patterns generalize beyond the task family studied here. Third, cross-architecture evaluation on non-Pi0-family VLAs (OpenVLA, RT-2) would determine the prevalence of fake safety in the broader VLA landscape. Fourth, we recommend that future VLA safety evaluations report both raw CAR and CAR_f; the gap between the two quantifies the degree of action-collapse contamination in a given model. Fifth, the catastrophic TSR collapse observed during fine-tuning (Section 6) motivates investigation into training strategies that preserve task capability while retaining collision avoidance learned from demonstrations.

Broader impact. Metric reliability is a prerequisite for safe deployment of autonomous manipulation systems. Contaminated evaluation metrics create a risk that models with no genuine collision avoidance capability pass safety benchmarks and proceed to deployment. Displacement filtering is a low-cost correction: it requires only end-effector position logging, no additional data collection or model modification. The analysis in this work is conducted entirely in simulation; real-world deployment introduces additional safety considerations (sensor noise, sim-to-real transfer, dynamic obstacles) that fall outside the present scope.

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A Displacement Threshold Sensitivity Analysis

To validate the robustness of our displacement threshold τ_d, we evaluate the frozen rate across four orders of magnitude. Table 6 shows that the fake safety pattern persists across all reasonable threshold choices: A1-LoRA exhibits near-total action collapse regardless of threshold, while Pi0.5+A1 maintains 0% frozen rate even at the most conservative setting.

Table 6: Frozen rate (% of episodes with $d_{\max} < \tau_d$) across threshold values.

τ_d	A1-LoRA (Pi0)	Pi0.5 Baseline	Pi0.5+A1 (SDF aux)
0.01 mm	[]	[]	[]
0.1 mm	[]	[]	[]
1.0 mm	[]	[]	[]
10 mm	[]	[]	[]

The choice of $\tau_d = 0.1$ mm represents a conservative threshold: end-effector displacement below this value is physically negligible in manipulation tasks where typical action magnitudes are 1–10 mm per step. Our conclusions hold for any threshold in the range $[0.01, 1.0]$ mm.

B Per-Task Breakdown

Table 7 provides the full per-task decomposition for the object_pudding suite (Level I). Task difficulty varies substantially: Task 1 (chocolate_pudding) consistently shows the lowest CAR across all methods due to tight obstacle placement, while Task 0 (orange_juice) shows the highest CAR.

Table 7: Per-task metrics on SafeLIBERO object_pudding Level I (50 episodes per task).

Task	Method	TSR (%)	CAR (%)	CAR_f (%)
T0: orange_juice	Pi0.5 Baseline	[]	[]	[]
	Pi0.5+A1 λ=0.005	[]	[]	[]
	Pi0.5+A1 λ=0.01	[]	[]	[]
T1: chocolate_pudding	Pi0.5 Baseline	[]	[]	[]
	Pi0.5+A1 λ=0.005	[]	[]	[]
	Pi0.5+A1 λ=0.01	[]	[]	[]
T2: milk	Pi0.5 Baseline	[]	[]	[]
	Pi0.5+A1 λ=0.005	[]	[]	[]
	Pi0.5+A1 λ=0.01	[]	[]	[]
T3: bbq_sauce	Pi0.5 Baseline	[]	[]	[]
	Pi0.5+A1 λ=0.005	[]	[]	[]
	Pi0.5+A1 λ=0.01	[]	[]	[]

C Training Details

Architecture. We use LoRA [Hu et al., 2022] adapters injected into the PaliGemma vision encoder (rank 16, α=16) and the action expert (rank 32, α=32), following the configuration of OpenPi. The SDF decoder is a 4-layer MLP (hidden dimension 256) with skip connections, taking pooled suffix features (1024-d) and query points (3-d) as input. The end-effector extractor is a 2-layer MLP ($1024 \to 256 \to 3$).

Training schedule. Training proceeds in two phases:

- **Phase 1** (steps 0–5000): Backbone and LoRA parameters are frozen. Only the SDF decoder and EE extractor are trained, allowing them to learn meaningful representations before the policy adapts.
- **Phase 2** (steps 5000–30000): LoRA parameters are unfrozen with a 100-step gradient warmup ramp. All auxiliary heads continue training jointly with the policy.

Hyperparameters.

Data. We use the SafeLIBERO object_pudding suite with pre-computed SDF ground truth. SDF values are computed offline from the obstacle mesh using a dense 64^3 voxel grid. During training, 64 query points are randomly sampled per step from this grid.

Computational resources. All training experiments are conducted on NVIDIA **[GPU model]** GPUs with **[XX]** GB memory. Phase 1 (5K steps) takes approximately **[X]** hours on a single GPU; Phase 2 (25K steps) takes approximately **[X]** hours. Evaluation of 200 episodes requires approximately **[X]** minutes. Total computational budget for the experiments reported in this paper is approximately **[X]** GPU-hours.

Table 8: Training hyperparameters.

Parameter	Value
Base model	Pi0.5 (OpenPi, $\sim$3.5B parameters)
Optimizer	AdamW
Learning rate	2.5×10^{-5} (cosine decay to lr/10)
Warmup steps	1000
Batch size	32
Total steps	30000
λ_{sdf}	$\{0.005, 0.01\}$
λ_{ee}	0.01
SDF query points per step	64
SDF clamp δ	0.05 m
Eikonal weight	0.1
LoRA rank (vision)	16
LoRA rank (action)	32
Phase 1 duration	5000 steps